# Advancing workforce pathways into the semiconductor manufacturing sector

## Specific Aims

Aim 1: Advance workforce pathways into the semiconductor manufacturing sector via a survey instrument-focused strategy.

Aim 2: Advance workforce pathways into the semiconductor manufacturing sector via a survey instrument-focused strategy.

Aim 3: Advance workforce pathways into the semiconductor manufacturing sector via a survey instrument-focused strategy.

## Background

This proposal addresses workforce pathways into the semiconductor manufacturing sector, a problem with substantial significance. Prior work has established a partial understanding, yet key gaps remain. Rigorous, pre-registered analyses will guard against bias and support reproducibility. We will assemble a multidisciplinary team with complementary expertise.

## Approach

Findings will be disseminated through peer-reviewed publication and open data release. The approach integrates established survey instrument methods with a novel analytical pipeline. Our preliminary data suggest a tractable path toward measurable improvement. This proposal addresses workforce pathways into the semiconductor manufacturing sector, a problem with substantial significance.

## Innovation

Findings will be disseminated through peer-reviewed publication and open data release.

## Investigator Context

Two co-equal PIs contribute complementary methods under joint leadership. The R2 host institution offers focused departmental resources and regional research partnerships, backed by a growing institutional research base.

## Budget Justification

Total requested support is $506,870 over 2 years at a R2 institution, covering personnel, materials, and dissemination.

## References


1. [1] https://doi.org/10.1371/journal.pone.0173664

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4. [4] https://doi.org/10.1126/science.1259855
5. [5] https://doi.org/10.1038/s41586-020-2649-2
6. [6] https://doi.org/10.1109/TPAMI.2016.2577031